



Identifying ion and carrier behavior in perovskite solar cells through impedance spectra



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Due to the interference of ion dynamics in perovskite materials, specifically the mixed ionic-electronic conduction property, the impedance response of perovskite solar cells exhibits numerous anomalous features. Herein, we conduct numerical simulations using the drift-diffusion model coupled with ion migration to elucidate how ionic and carrier behavior govern the impedance response. Here we show that, in the Nyquist plot, the low frequency and high frequency semicircular features of direct current voltage-dependent impedance spectroscopy are determined by ion migration and carrier transport, respectively. The low-frequency semicircular feature disappears when ion migration is fully suppressed. As ion mobility or mobile ion concentration increases, both the low-frequency decay region of the real component and low-frequency peak of the imaginary component in frequency-domain impedance plot shift toward higher frequencies. As the bulk or interface carrier recombination rate increases, both the high frequency decay region of the real component and the high frequency peak of the imaginary component shift toward higher frequencies.

Due to the advantages of a simple preparation process, low cost, and significant potential for performance improvement, perovskite solar cells (PSCs) have emerged as one of the most intensively studied topics in photovoltaic (PV) field^{1–6}, with the current record for power conversion efficiency (PCE) exceeding 27%⁷. To further enhance the PCE and stability of PSCs, it is crucial to passivate detrimental defects that act as recombination centers, thereby reducing energy loss from nonradiative recombination, and to suppress ion migration to prevent phase transition or crystal destruction of perovskite materials^{8–12}. In this context, the critical step is to accurately quantify nonradiative recombination and ion migration in PSCs.

Impedance spectroscopy (IS) is a straightforward and versatile characterization technique widely used to extract photoelectrical parameters of PV devices, such as carrier diffusion coefficient, recombination lifetime, carrier diffusion length, carrier mobility, trap state distribution, and ion diffusion coefficient^{13–16}. Unlike many other characterization methods that require device destruction or are performed on half cells^{17–19}, IS can be conducted on complete devices under operating conditions^{12–15}. Given these advantages, Bisquert and co-workers developed a comprehensive theory for extracting carrier dynamics in dye-sensitized solar cells based on equivalent circuit models^{13,14,20}. However, the involvement of ionic motion complicates

the physical mechanism of PSCs, making it distinct from dye-sensitized solar cells^{13,15,21,22}. This complexity poses challenges in extracting physical parameters from PSCs using their approach. Although researchers have attempted to extend this method to PSCs, they have encountered difficulties in providing meaningful interpretations for equivalent circuit elements (such as resistances and capacitances), even when the IS results can be well fitted to such models^{13,15,23}. Riquelme et al. proposed an effective strategy to circumvent the limitations of equivalent circuit approaches: they performed numerical simulations of PSCs based on a drift-diffusion model that incorporates ion motion, achieving good agreement with experimental IS results²⁴. Notably, all parameters extracted from the drift-diffusion model possess clear physical meanings, unlike those from equivalent circuit models. Subsequently, Bennett et al. derived analytical equations to describe the impedance response of PSCs²⁵, which further advances the understanding of the underlying impedance mechanisms and provides a more convenient approach for parameter extraction. Based on the drift-diffusion model coupled with ion migration, Clarke et al. elucidated the physical origins corresponding to six types of Nyquist spectral features of PSCs²⁶. Despite these breakthroughs in understanding the impedance response of PSCs, how to identify between ion and carrier behavior in PSCs based on IS

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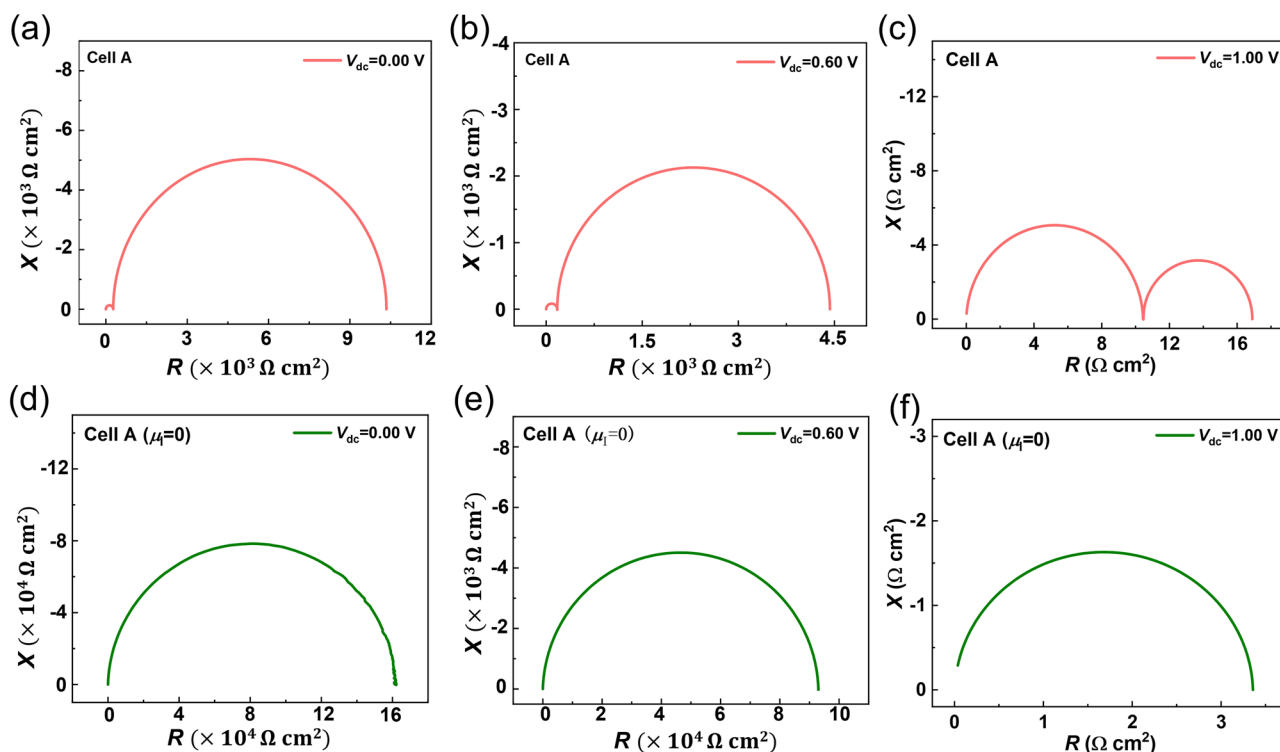


Fig. 1 | Impedance spectra in Nyquist diagrams of Cell A simulated under different direct current voltage for the cases with or without ion migration. The applied direct current voltages are **a, d** 0 V, **b, e** 0.6 V, and **c, f** 1 V. Here, R and X

represent the real and imaginary components of the impedance, respectively. V_{dc} represents the direct current voltage. μ_i represents the mobility of mobile ion.

measurement remains elusive, and the quantitative influence of key device parameters, such as ion mobility, mobile ion concentration, and carrier recombination rate, on the IS response still requires further analysis.

In this work, we systematically investigate the extraction of ion migration and carrier recombination information from IS data of PSCs, focusing on the following questions: (i) How can ion migration and carrier recombination parameters be extracted from impedance responses? (ii) What role does ion migration play in the impedance response? (iii) How does the impedance response vary with ionic charge mobility or carrier recombination rate? To address the above issues, we perform numerical simulations of the impedance response of PSCs based on a drift-diffusion model that incorporates ion migration. We find that the low-frequency region of the impedance response is governed by ion migration, while carrier transport mainly affects the high-frequency region. When ion migration in the perovskite active layer is completely suppressed, the low-frequency arc in the impedance response disappears. The low-frequency decay in the real component and the low-frequency peak in the imaginary component of the IS shift toward higher frequencies as ion mobility or mobile ion concentration increases. Similarly, the high-frequency decay in the real component and the high-frequency peak in the imaginary component shift toward higher frequencies as the bulk or interface carrier recombination rate increases. These results offer valuable guidance for extracting ion migration and carrier recombination parameters from IS experimental data.

Results and discussion

Ion migration in impedance response

Taking the device with normal hysteresis (cell A) as an example for investigation, the simulated J - V sweep curves of cell A at scan rates of 0.0015 Vs^{-1} , 0.015 Vs^{-1} , 0.15 Vs^{-1} , 1.5 Vs^{-1} , 15 Vs^{-1} and 150 Vs^{-1} are shown in Fig. S2a–f, respectively. The ionic charge motion causes changes in the charge accumulation within the Debye layers in the perovskite near both interfaces. This leads to different electric fields for the same applied voltage values during

reverse and forward sweeps, hence causing hysteresis^{27–30}. Accordingly, hysteresis can only be observed when ion mobility matches the scan rate, which results in scan rate-dependent hysteresis. Figure 1a–c shows the simulated impedance response represented in Nyquist plots at a series of applied voltages. As the direct current voltage (dc voltage) increases, the overall device impedance decreases dramatically, and the difference in radius between the low-frequency arc and the high-frequency arc also decreases dramatically. From the frequency response shown in Fig. S3, it can be seen that with increasing dc voltage, the low-frequency decay region of the real component and the low-frequency peak of the imaginary component remain unchanged in the frequency region from 10^{-4} to 1 Hz, while the high-frequency decay region of the real component and the high-frequency peak of the imaginary component shift toward higher frequencies. This finding is consistent with the results reported by Bennett et al.²⁵.

To explore the role that ionic charge motion plays in impedance response, the mobility of ionic charge in cell A is set to zero to eliminate the interference of ion migration. Expectedly, when the ion migration in cell A is completely suppressed, no hysteresis could be observed in all of the J - V sweeps (Fig. S4). Moreover, all the J - V curves are identical and independent of scan rate. Regarding the impedance response, as shown in Fig. 1d–f, a semicircular feature in the low-frequency region disappeared, indicating that this low-frequency semicircular feature is attributed to ion migration. As the dc voltage increases, the decay region of the real component and the peak of the imaginary component in the impedance response frequency plot shift toward higher frequencies (Fig. S5), indicating a higher-frequency behavior compared to the case with ion migration. Under this condition, only the high-frequency decay region of the real component and the high-frequency peak of the imaginary component can be observed in the simulated R -frequency and X -frequency plots, whereas the corresponding low-frequency features vanish. This indicates that the low-frequency features originate from ion migration. Furthermore, under such circumstances, only carrier transport occurs in the device, so the high-frequency features are caused by carrier behavior.

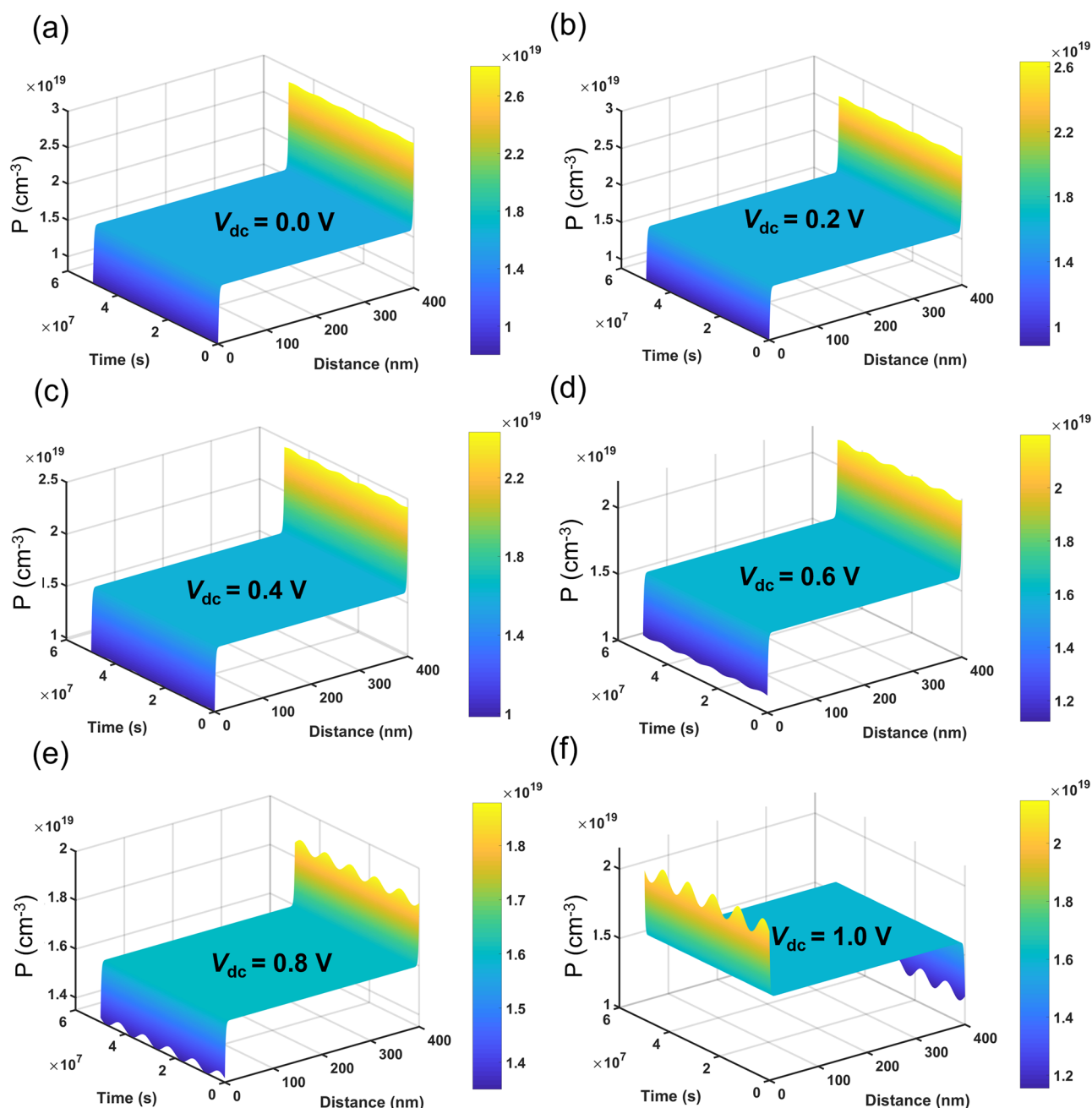


Fig. 2 | Evolution of ion distribution in the perovskite active layer of cell A during impedance spectra simulation at a low frequency of 10^{-7} Hz. The applied direct current voltages are a 0 V, b 0.2 V, c 0.4 V, d 0.6 V, e 0.8 V, and f 1 V. Herein, P

denotes the distribution of mobile ion concentration in the perovskite active layer. V_{dc} represents the direct current voltage.

To explore the ion migration process during the impedance response simulation, the evolution of the ion distribution in the perovskite active layer at two extreme frequencies (10^{-7} Hz and 10^7 Hz) is provided in Fig. 2 and S6, 7. Under the combined influence of built-in voltage and applied voltage, ions migrate between two regions known as Debye layers adjacent to the ETL/PAL and PAL/HTL interfaces^{27,28,31,32}. The resulting ion distribution is governed by the interplay between the built-in voltage and the applied voltage. When the applied dc voltage is less than the built-in voltage (Figs. 2a–e and S6a–e), ionic charges accumulate in the Debye layer adjacent to the PAL/HTL interface. When the applied dc voltage equals the built-in voltage, the ionic charges become uniformly distributed in the perovskite active layer (Fig. S7). Conversely, when the dc voltage exceeds the built-in voltage (Fig. 2f and S6f), ionic charges accumulate in the Debye layer adjacent to the PAL/ETL interface. A low-frequency additional small

sinusoidal voltage perturbation induces back-and-forth motion of ionic charges between the Debye layers, leading to slight sinusoidal fluctuations in ionic charge concentration (see Fig. 2 and Fig. S7a). In contrast, under high-frequency perturbation, insufficient time is available for ionic charges to traverse from one Debye layer to the other, preventing sinusoidal fluctuation of ionic charge concentration (Figs. S6, 7b).

Taking the device with inverted hysteresis (cell B) as an example for further investigation, Fig. S8 presents the J - V curves of cell B at the scan rates of 0.0015 V s^{-1} , 0.015 V s^{-1} , 0.15 V s^{-1} , 1.5 V s^{-1} , 15 V s^{-1} and 150 V s^{-1} . Similarly, in cell B, both the impedance value and the difference in radius between the two semicircular features decrease significantly with increasing dc voltage (see Fig. 3a–c). Notably, the imaginary component of the low-frequency semicircular feature becomes negative when the dc voltage reaches 1 V (see Fig. 3c), a behavior consistent with the experiment results reported by

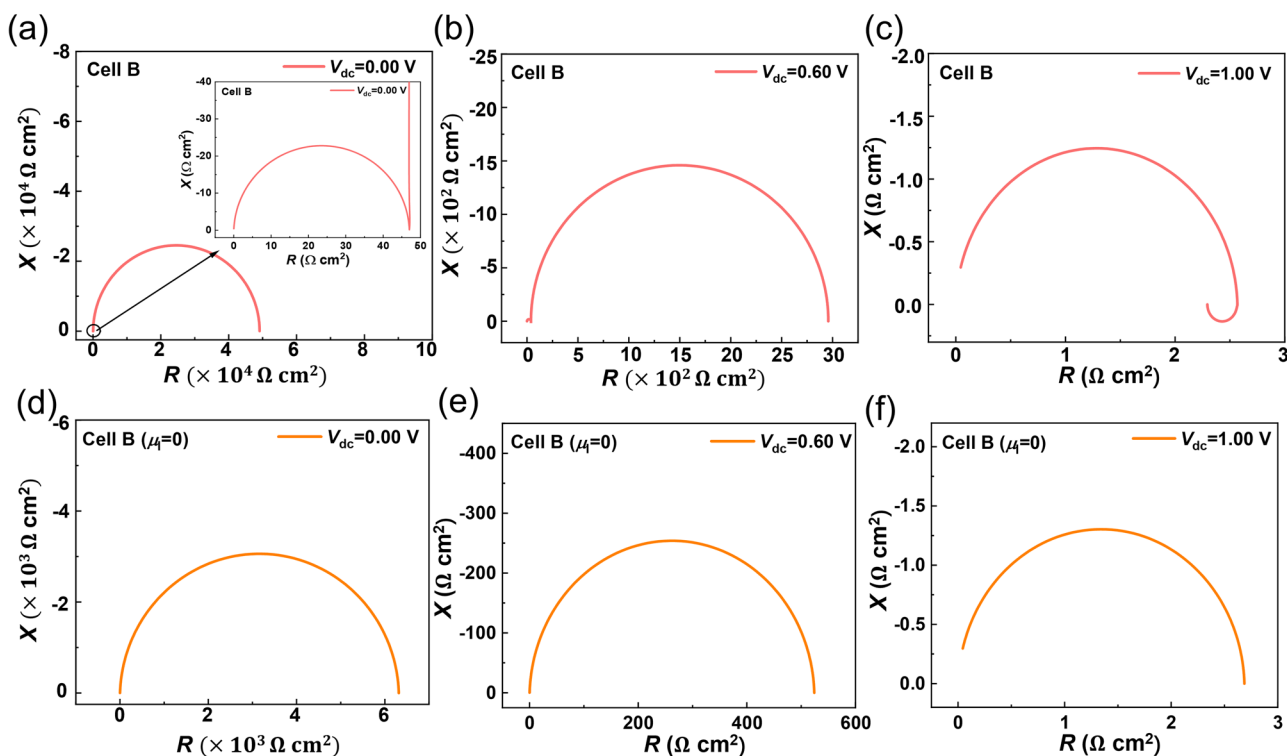


Fig. 3 | Impedance spectra in Nyquist diagrams of Cell B simulated under different direct current voltage for the cases with or without ion migration. The applied direct current voltages are **a, d** 0 V, **b, e** 0.6 V, and **c, f** 1 V. Here, R and X

represent the real and imaginary components of the impedance, respectively. V_{dc} represents the direct current voltage. μ_1 represents the mobility of mobile ion.

Bisquert et al.³³. Their study demonstrated that only the inverted perovskite device shows negative capacitance at specific applied dc voltages. Given that the device type (normal or inverted PSCs) is governed by carrier recombination, a correlation exists between negative capacitance and carrier recombination. The low-frequency negative capacitance observed in perovskite solar cells arises from the dynamic coupling between slow ionic polarization and interfacial carrier recombination. Under external direct-current bias, mobile halide vacancies migrate and accumulate space charge at the film interfaces, which continuously modulates the interfacial electric field. The time lag between ionic relaxation and electronic recombination accelerates the Shockley–Read–Hall non-radiative recombination of photogenerated carriers, thereby generating an inductive response that manifests as negative capacitance on impedance Nyquist plots³⁴. Moreover, with increasing dc voltage, the high-frequency decay region of the real component and the high-frequency peak of the imaginary component shift toward higher frequencies (Fig. S9). When ion migration in cell B is completely suppressed, all J - V curves overlap completely and exhibit no hysteresis (Fig. S10). In the corresponding impedance spectra, the low-frequency semicircular arc associated with ion migration disappears (Fig. 3d–f). Our simulations further confirm this mechanism: negative capacitance disappears completely once we turn off the ionic modulation effect on carrier recombination. Only when ion migration and interfacial recombination work together can the negative capacitance be reproduced in numerical simulations. Additionally, similar to Cell A under fully suppressed ion migration, with increasing dc voltage, the decay region in the real component and the peak in the imaginary component shift toward higher frequencies (Fig. S11). The evolution of ionic distribution in the perovskite active layer of Cell B during impedance simulations is similar to that of Cell A (Figs. S12–14).

Carrier recombination in impedance response

An analytical model of IS for PSCs is employed to investigate carrier recombination based on the impedance response of the device. The details of

the analytical model are provided in Supplementary Note 3 of the Supplementary Information. Figure 4a shows the recombination current density for four devices at different applied dc voltages. The recombination current density is primarily governed by carrier dynamics, whereas ion migration has a relatively small influence. The recombination current of cell B is significantly higher than that of cell A, indicating that the carrier recombination loss in cell B is substantially greater than in cell A. This is attributed to the higher bulk and interface carrier recombination parameters set for cell B (see Table S2). The electronic ideality factor (used to describe carrier recombination) shown in Fig. 4b also suggests a higher carrier recombination rate in cell B. Figure 4c presents the apparent ideality factor (used to describe both carrier recombination and ion migration) for the four devices. Figure 4d shows the difference between the apparent and electronic ideality factors for the four devices. According to eq. (S15) and eq. (S17), the value of $n_{ap} - n_{el}$ for cell B is negative at 0.96 V and 1 V, accounting for the negative low-frequency resistance and capacitance observed for cell B at these two dc voltages (Table S8). Since the value of $n_{ap} - n_{el}$ for cell A ($\mu_1 = 0$) and B ($\mu_1 = 0$) is zero, the low-frequency resistance for these devices is also zero (Tables S5 and S6).

Based on the RC-RC equivalent circuit (Fig. S1 in the Supplementary Note 4 of Supplementary Information) used to model the impedance response of PSCs, the relevant parameters describing device properties are extracted. As shown in Fig. 5a, the low-frequency resistance of cell A is significantly higher than that of cell B in the dc voltage range from 0.84 V to 1 V. As the dc voltage increases, the low-frequency resistance decreases for both cell A and cell B decreases. For high-frequency resistance (Fig. 5b), with increasing dc voltage, the high-frequency resistance also decreases across all four devices. Regarding the low-frequency capacitance (Fig. 5c), the values for cell A and cell B are similar in the dc voltage range from 0.84 V to 0.92 V. The value for cell A remains positive over the entire dc voltage range, whereas for cell B, it becomes negative in the range from 0.92 V to 1 V. For the high-frequency capacitance shown in Fig. 5d, the values for all four cells remains nearly constant with increasing dc voltage, which is consistent with

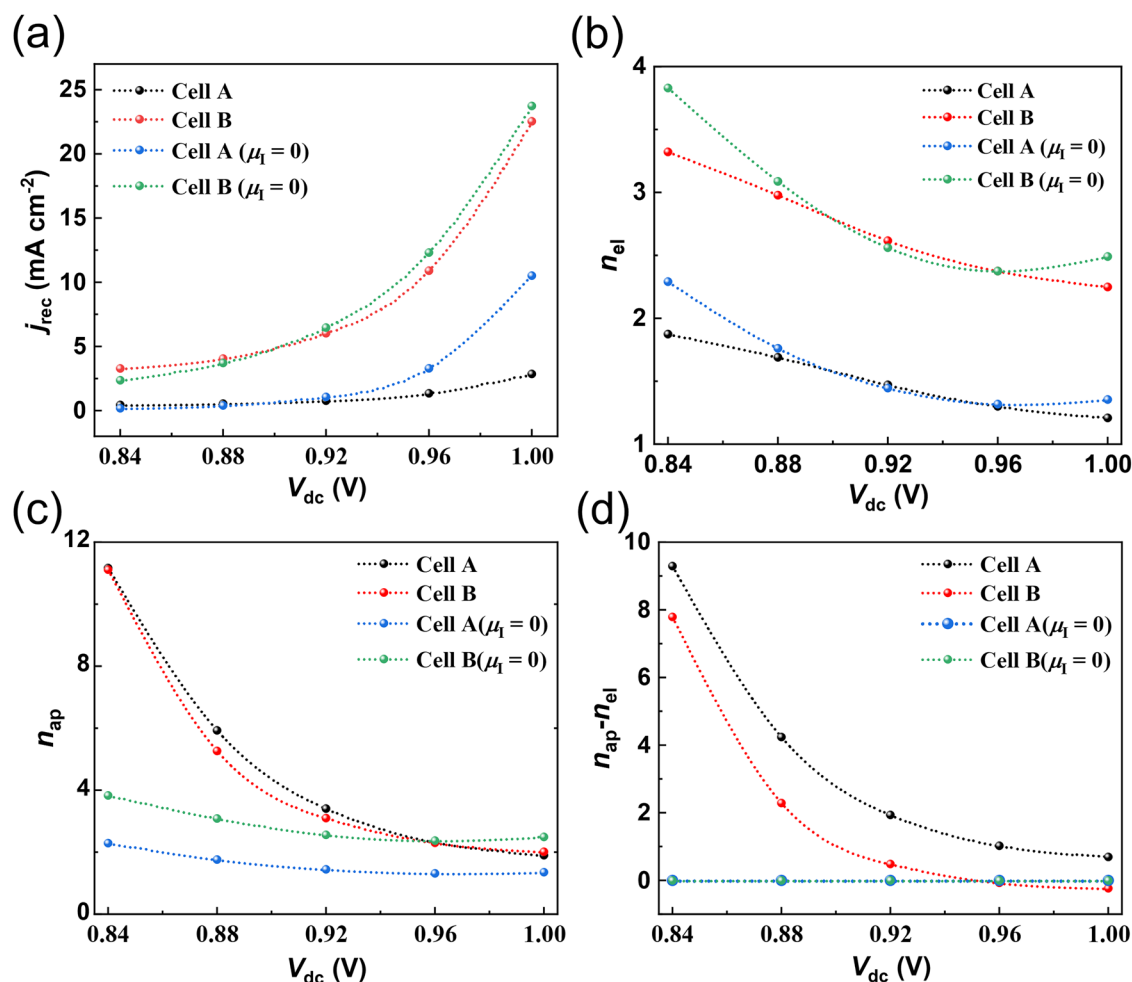


Fig. 4 | Four physical quantities extracted from impedance spectra using the analytical model. **a** Recombination current, **b** electronic ideality factor, **c** apparent ideality factor, and **d** the difference between apparent and electronic ideality factors for four devices (cell A, cell B, Cell A ($\mu_i = 0$), Cell B ($\mu_i = 0$)) extracted from the impedance spectroscopy. Here, V_{dc} is the direct current voltage, j_{rec} is the

recombination current density, n_{ap} is the apparent ideality factor, and n_{el} is the electronic ideality factors. μ_i represents the mobility of mobile ion. All values for the above results are provided in the Table S7-S10 (Supplementary Note 6 of Supplementary Information).

the results (blue solid line) calculated using the analytical formula of equation S3.

Effect of ion mobility and concentration on impedance response

Figure 6a–f presents the frequency plots of impedance spectra for cell A, simulated by varying the parameters that determine ion mobility, including the ion migration energy barrier E_a , high-temperature ion diffusion coefficient D_{inf} , and ambient temperature T^{85-37} . As ion mobility increases (i.e., with decreasing ion migration energy barrier E_a , or increasing high-temperature ion diffusion coefficient D_{inf} or increasing ambient temperature T), the low-frequency decay region of the real component and the low-frequency peak of the imaginary component both shift toward higher frequencies. In contrast, the high-frequency decay region of the real component and the high-frequency peak of the imaginary component are governed by carrier transport and are independent of ion migration. Therefore, they remain unchanged with increasing ion mobility (Fig. 6a, b, d, e). Notably, when the ambient temperature T increases, carrier mobility also increases. Thus, in the shift toward higher frequencies of high-frequency features observed with increasing temperature is attributed to the increase in carrier mobility, rather than ion mobility (see Fig. 6c, f). These trends are also fully applicable to cell B (Fig. S15).

The concentration of mobile ions varies among different perovskite materials, making it important to investigate its influence on the device

impedance response. Figure 6g–j displays the impedance response spectra for cell A and cell B at different mobile ion concentrations. An increase in mobile ion concentration is equivalent to an increase in ion vacancy mobility. Accordingly, for both cell A and cell B, as the mobile ion concentration increases, the low-frequency decay region of the real component and the low-frequency peak of the imaginary component shift toward higher frequencies, while the high-frequency features remain unchanged.

Effect of recombination rate on impedance response

Figure 7 presents the impedance responses for cell A and cell B simulated by varying the bulk recombination parameters of τ_n and τ_p . The radius of both low-frequency and high-frequency semicircular arcs increases as the carrier recombination rate decreases (i.e., as τ_n and τ_p increase). Therefore, a larger semicircular arc radius in the impedance response indicates a lower bulk defect concentration in the device (see Figs. 7a–d). As the carrier recombination rate increases (i.e., τ_n and τ_p decrease), the high-frequency decay region of the real component and the high-frequency peak of the imaginary component shift toward higher frequencies (Fig. 7b, c, e, f). In contrast, the low-frequency decay region of the real component and the low-frequency peak of the imaginary component are governed by ion migration and are independent of carrier transport. Thus, they remain unchanged with increasing carrier recombination rate (Fig. 7b, c, e, f). The same trend

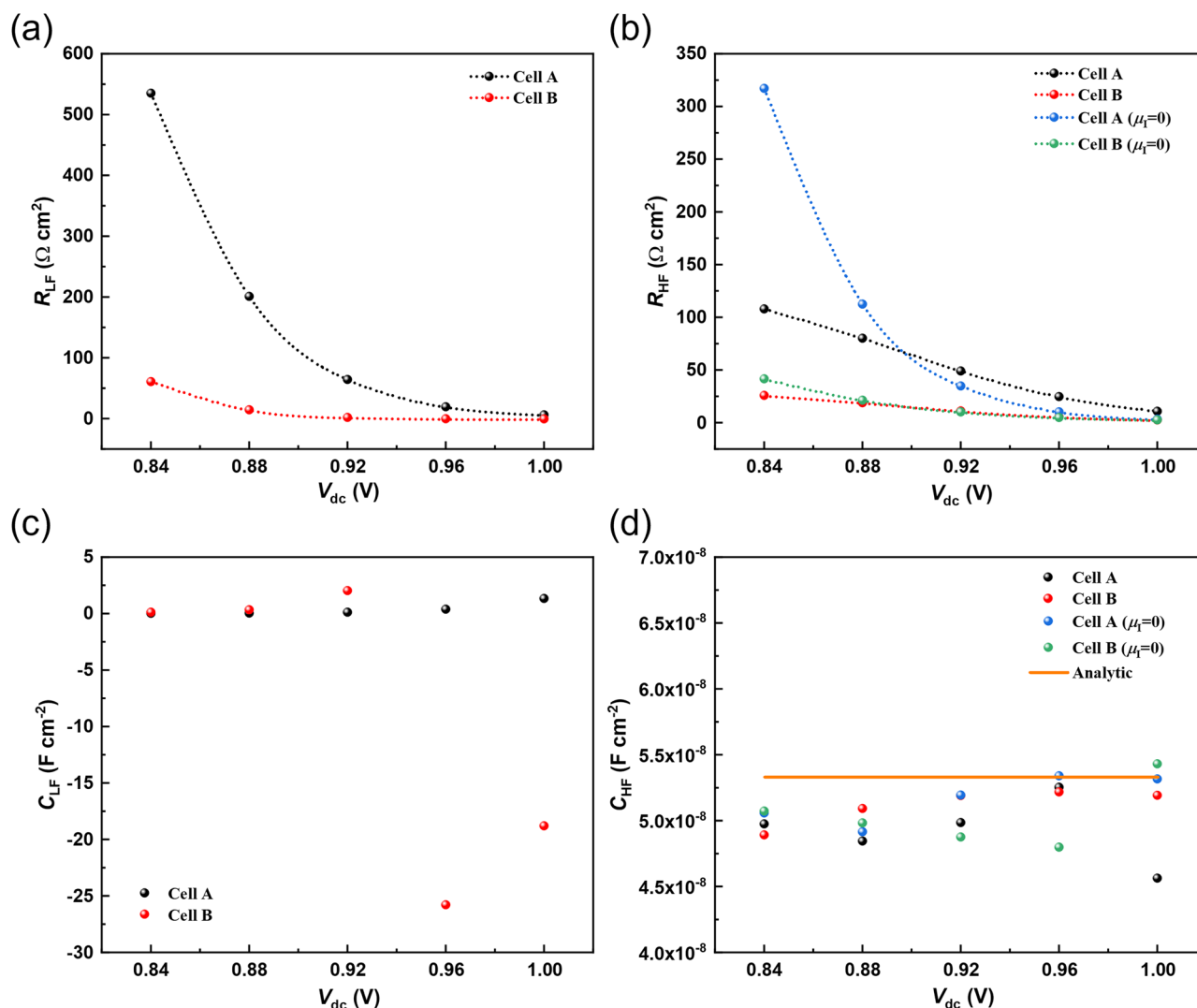


Fig. 5 | Resistances and capacitances extracted from impedance spectra using the RC-RC equivalent circuit model. **a** Low-frequency resistance, **b** high-frequency resistance, **c** low-frequency capacitance, and **d** high-frequency capacitance for the four devices (cell A, cell B, Cell A ($\mu_i = 0$), Cell B ($\mu_i = 0$)) extracted from the impedance spectra based on the RC-RC equivalent circuit. Here, V_{dc} represents the

direct current voltage, R_{LF} is the low-frequency resistance, R_{HF} is the high-frequency resistance, C_{LF} is the low-frequency capacitance, C_{HF} is the high-frequency capacitance. μ_i represents the mobility of mobile ion. All values for the above results are provided in the Table S7–10 (Supplementary Note 6 of Supplementary Information).

observed for the high-frequency semicircular arcs in cell A ($\mu_i = 0$) and cell B ($\mu_i = 0$) is also confirmed (Fig. S16).

Figure 8 shows the impedance responses of cell A and cell B simulated by varying the interface recombination parameters. The corresponding interface recombination parameters are provided in Table S3 and S4 (Supplementary Note 1 of Supplementary Information). Similarly, the radius of both the low-frequency and high-frequency semicircular arc increases as the interface recombination rate decreases (Fig. 8a, d), suggesting that a larger radius corresponds to a lower interface defect concentration in the device. Since the low-frequency decay region of the real component and the low-frequency peak of the imaginary component are unrelated to carrier transport, they remain unchanged with increasing interface recombination rate (Fig. 8b, c, e, f). Meanwhile, the high-frequency decay region of the real component and the high-frequency peak of the imaginary component shift toward higher frequencies as the interface recombination rate increases (Fig. 8b, c, e, f). As shown in Fig. S17, a consistent trend is observed for the high-frequency semicircular arcs in both cell A ($\mu_i = 0$) and cell B ($\mu_i = 0$). The corresponding interface recombination parameters are provided in Table S5 and S6 (Supplementary Note 1 of Supplementary Information).

limitations of this model

It is worth noting that there are several inevitable limitations in the present numerical simulation model. First, this work adopts an ideal homogeneous perovskite layer without considering the actual grain boundary effect, which acts as a major ion migration channel and defect recombination center in real polycrystalline perovskite devices. Second, the established model only focuses on single-halide perovskite systems, while mixed halide components and corresponding halide segregation effects under electrical bias are not incorporated in the simulation. Third, temperature-induced crystal-phase instability and irreversible phase transitions in perovskite materials are neglected in this quasi-static impedance simulation. In addition, this study adopts uniform, idealized defect-distribution settings for numerical calculations, rather than the random, uneven defect distribution observed in actual fabricated perovskite solar cells. The above neglected factors will further affect the hysteresis behavior and impedance response of real devices, and can be incorporated into future optimized models.

Conclusion

In summary, numerical simulations of PSC impedance spectra under varied carrier recombination and ion migration parameters allow us to distinguish

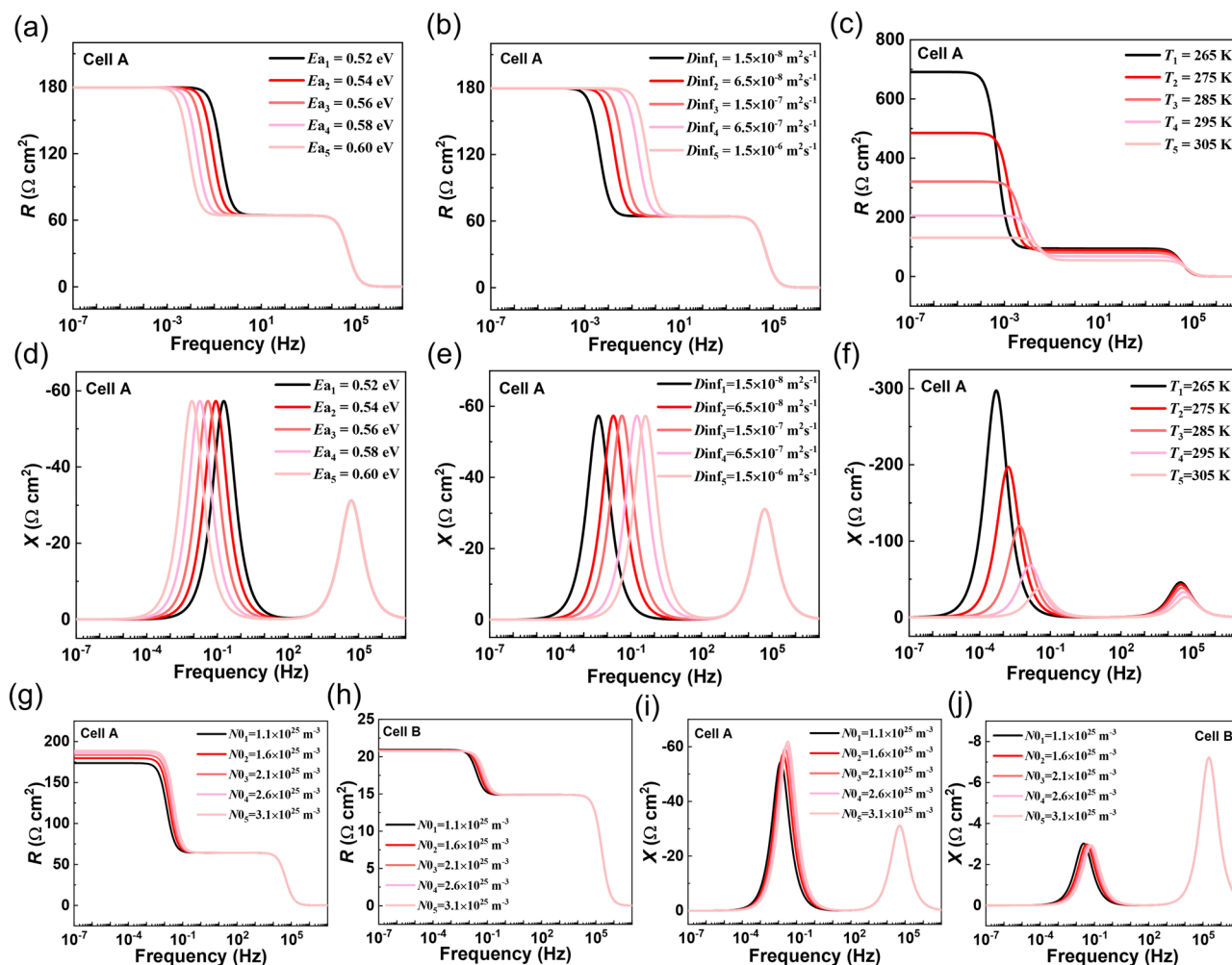


Fig. 6 | Frequency plots of impedance spectra for cell A and cell B simulated by varying the parameters that determine ion mobility and mobile ion concentration. Tuned parameters for Cell A: a, d ion migration energy barrier E_a , b, e high temperature ion diffusion coefficient D_{inf} , c, f ambient temperature T , g, i mobile ion

concentration N_0 ; tuned parameter for Cell B: h, j mobile ion concentration N_0 . Here, R and X represent the real and imaginary components of the impedance, respectively.

the separate contributions of ionic migration and carrier recombination of PSCs. It is shown that the low- and high-frequency semicircles in IS are governed by ion and carrier dynamics, respectively. The low-frequency semicircle disappears when ion migration in the perovskite layer is completely suppressed. The occurrence of negative capacitance originates from carrier recombination mechanisms coupled with ion migration. As ion mobility or mobile ion concentration increases, the low-frequency decay region of the real component and the low-frequency peak of the imaginary component of the impedance response shift toward higher frequencies, while the high-frequency semicircle remains unchanged. Similarly, with an increase in carrier recombination rate for both bulk and interface recombination, the high-frequency decay region of the real component and the high-frequency peak of the imaginary component of impedance response shift toward higher frequencies, while the low-frequency semicircle remain unchanged. The findings of this work advance the fundamental understanding of impedance behaviors in perovskite solar cells and provide theoretical guidance for quantifying ion migration and trap-assisted carrier nonradiative recombination in PSCs.

Methods

In this work, the IonMonger 2.0 software, based on the drift-diffusion model coupled with ion migration is adopted to carry out device simulation research^{31,32}. Clarke et al. systematically validated the IonMonger software

by qualitatively matching simulated impedance spectra with experimental measurements²⁶. The detailed introduction of the drift-diffusion model coupled with ion migration is provided in Supplementary Note 2 of the Supplementary Information. A planar standard PSC structure of fluorine-doped tin oxide (FTO)/electron transport layer (ETL)/perovskite active layer (PAL)/hole transport layer (HTL)/electrode is adopted. The finite element method is used to solve the spatial and temporal distribution of ion, electron, hole density, and electric potential within the device, based on a complete set of drift-diffusion equations coupled with Poisson's equation. The carrier generation rate follows the Beer-Lambert law under illumination equivalent to 1 sun. Various carrier recombination mechanisms, including radiative recombination, bulk and interface trap-assisted Shockley-Read-Hall recombination, and Auger recombination, are considered^{27–30,38–40}.

The key parameters are listed in Table S1 (Supplementary Note 1 of Supplementary Information), while the carrier recombination parameters are provided in Table S2 (Supplementary Note 1 of Supplementary Information). All simulation parameters in Tables S1 and S2 are extracted from experimental measurements and first-principles calculations, which have been widely adopted for device simulations of perovskite solar cells^{27–30}. Hysteresis type (normal or inverted) is determined by the bulk and interface carrier recombination rates. Two sets of recombination parameters, corresponding to normal hysteresis (cell A) and inverted hysteresis (cell B), are used in the simulations, as detailed in Table S2.

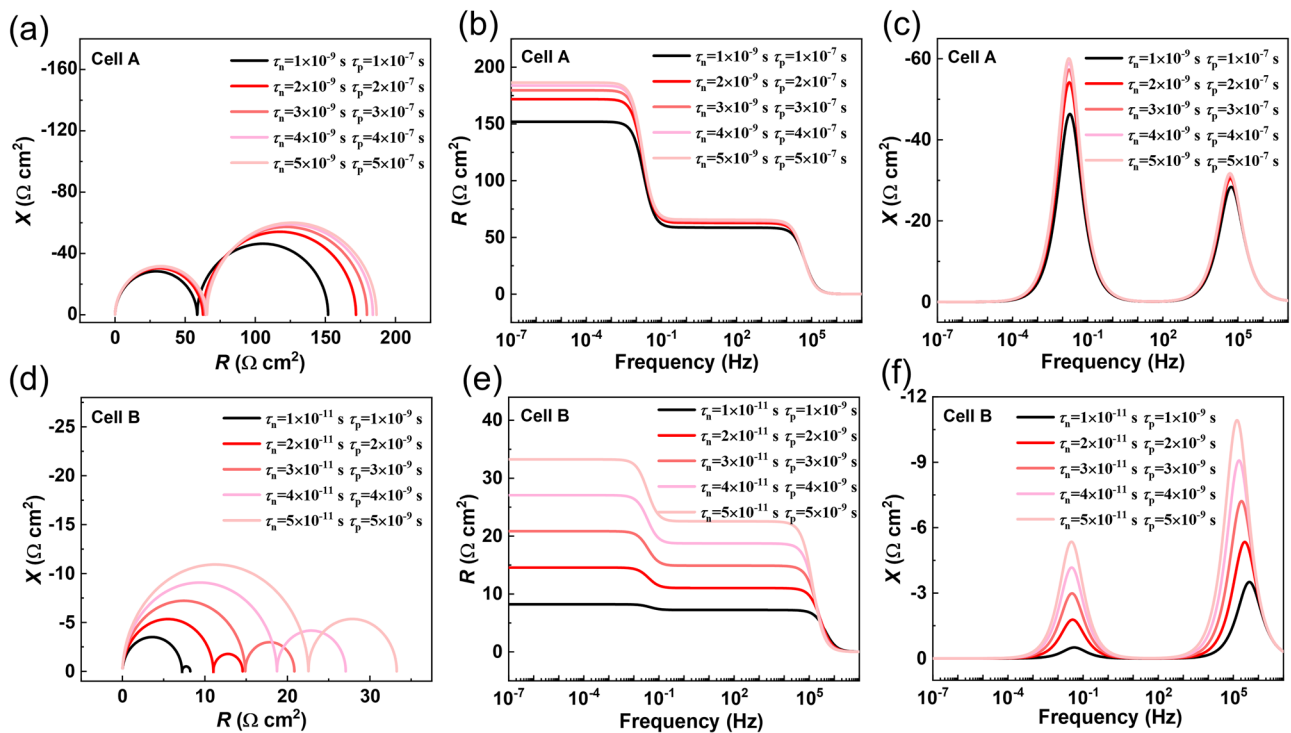


Fig. 7 | Impedance responses for cell A and cell B simulated by varying the bulk recombination parameters of electron pseudo-lifetime and hole pseudo-lifetime in perovskite. The results are obtained by tuning τ_n and τ_p : a–c cell A and d–f cell B.

Here, R and X represent the real and imaginary components of the impedance, respectively. τ_n and τ_p represent the electron pseudo-lifetime and hole pseudo-lifetime in perovskite, respectively.

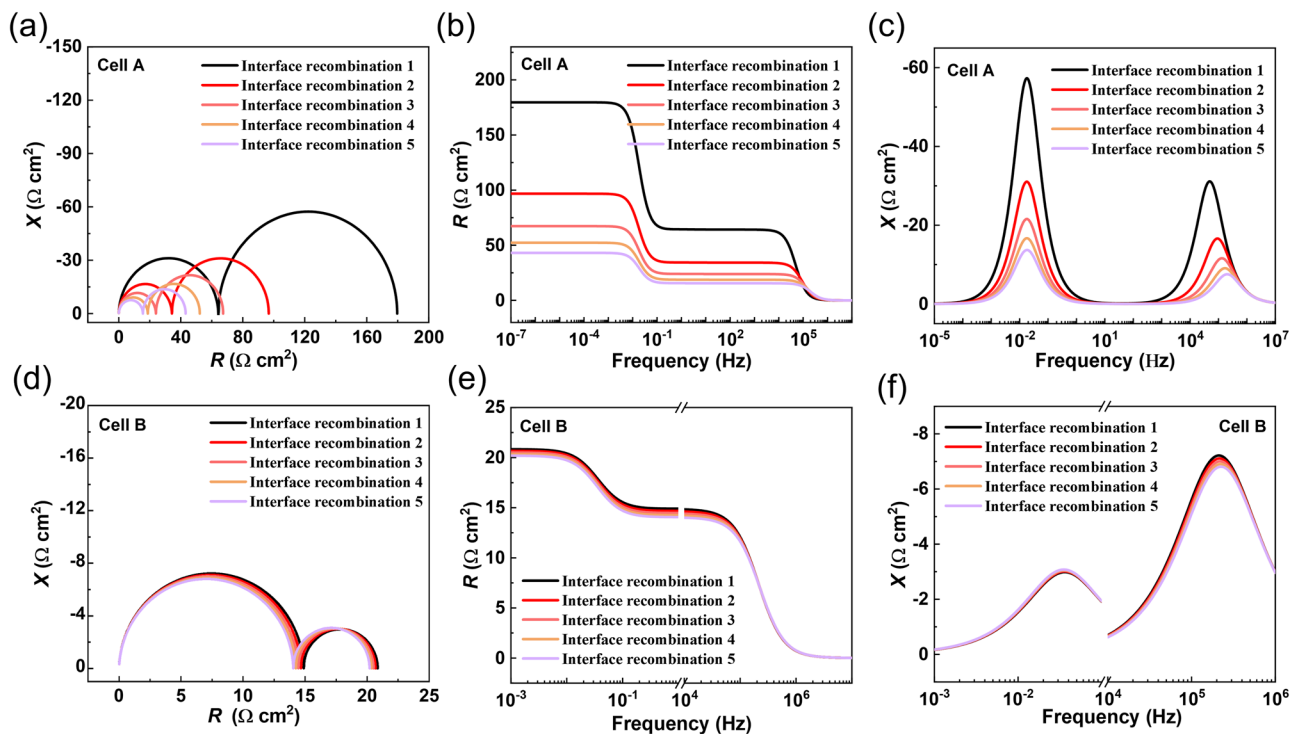


Fig. 8 | Impedance responses for cell A and cell B simulated by varying the interface recombination parameters. a–c cell A and d–f cell B. Here, R and X represent the real and imaginary components of the impedance, respectively.

In the current-voltage (J - V) sweep simulations, the applied voltage is initially preconditioned using a hyperbolic tangent function for 5 s. Subsequently, it swept linearly from 1.2 V to 0 V and then back to 1.2 V at specified bias scan rates.

For impedance response simulations, the applied voltage is defined as

$$V_{\text{applied}}(t) = V_{\text{dc}} + V_p \cos(\omega t) \quad (1)$$

where V_{dc} represents the direct current voltage (dc voltage). $V_p \cos(\omega t)$ denotes a small sinusoidal perturbation with amplitude V_p and frequency ω .

Impedance spectra $Z(\omega)$ can be obtained by simulating the device's response to small voltage perturbations at various frequencies under a fixed dc voltage. Herein, we adopt R and X to denote the real and imaginary parts of complex impedance, where $R = R(\omega) = \text{Re}(Z(\omega))$ and $X = X(\omega) = \text{Im}(Z(\omega))$.

Data availability

All data supporting the findings of this study are available from the corresponding authors upon request.

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Author contributions

Z.L. conceived the idea. Z.L. and Z.C. performed the simulation; Z.L. wrote the manuscript; All authors discussed the results and commented on the manuscript.

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Competing interests

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Additional information

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